

Quantum Walk Sampling for Extremely Rare Events Project Report

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Abstract

Estimating extremely small probabilities is important in reliability analysis, climate modeling, and astrophysics. Classical Monte Carlo becomes inefficient in the ultra-rare regime because observing a single event may require a very large number of samples. We examine whether quantum algorithms can improve rare-event probability estimation using two problem representations: a one-dimensional first-passage process with an absorbing boundary and a branching fault-tree model. Both models use quantum circuits with a flag qubit marking the target event. Amplitude estimation can estimate the flagged probability, provided the circuit coherently encodes the intended probability distribution. Experiments on small toy models illustrate the implementation and its resource requirements. Further validation is needed to establish its accuracy and potential advantage for extremely rare events.

1 Introduction

Many simulation methods perform well for common events but become slow and computationally expensive for extremely rare events arising in applications such as equipment reliability analysis, nucleosynthesis modeling, climate modeling, and cosmic-ray simulations. In the far tail of a distribution, many standard deviations from the mean, classical Monte Carlo becomes sample-inefficient and costly, often requiring millions or billions of samples to observe a single event.

To address this limitation, researchers use variance-reduction methods such as importance sampling, subset simulation, and splitting. Importance sampling modifies the sampling distribution so that rare events occur more frequently and then reweights the samples to preserve unbiased estimates. Subset simulation and splitting approach the rare region in stages, repeatedly propagating trajectories that reach intermediate thresholds toward the target region [1]. While methods can be effective, they require problem-specific tuning (intermediate thresholds, branching rules) and can struggle when the rare region has high-dimensional irregular shape, or multiple modes. In Markov-chain samplers, umbrella sampling, tempering, and surrogate modeling provide additional tools; however, they share a similar sensitivity to design choices.

2 Methods

We focus on the ultra-rare regime where the probability of interest lies in the range

$$p_T \in [10^{-12}, 10^{-6}].$$

Two problem representations are considered. The first is a one-dimensional first-passage process with an absorbing boundary. The second is a branching structure motivated by fault-tree models used in reliability analysis. In both cases we design quantum circuits based on discrete-time quantum walks and estimate the final probability using quantum amplitude estimation.

2.1 One-dimensional first-passage model

We consider a discrete stochastic process on positions

$$\{0, \dots, D\},$$

where the walker starts at position $X_0 = 0$ and evolves in discrete steps. At each step the walker moves forward or backward with fixed probability. The state D represents an absorbing boundary corresponding to the rare event of interest. Let

$$\tau_A = \inf\{t \geq 0 : X_t \in A\}, \quad A = \{D\},$$

denote the first hitting time of the absorbing set. The probability of interest is

$$p_T = \mathbb{P}(\tau_A \leq T),$$

which measures the probability that the walker reaches the absorbing boundary within T steps.

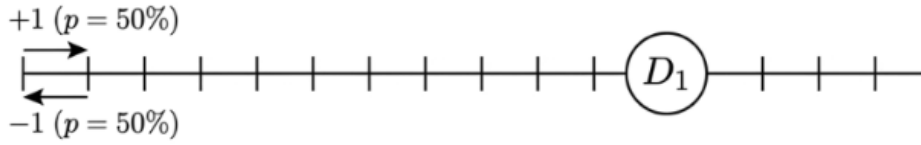


Figure 1: One-dimensional random walk with absorbing boundary D .

2.2 Quantum walk circuit design

We implement a discrete-time coined quantum walk, whose probability distribution generally differs from that of the classical process because of quantum interference. The position of the walker is encoded in a quantum register representing the states $\{0, \dots, D\}$, and a single qubit acts as the coin controlling the direction of motion.

Each step of the walk applies two operations:

- a Hadamard coin flip that creates a superposition of directions,
- a controlled shift that updates the position register conditioned on the coin state.

Reflecting and absorbing boundaries are implemented using reversible logic so that the walker cannot move below position 0, while state D represents the absorbing boundary. After T steps, a flag qubit is set to 1 if the walker reaches position D . Because the boundary is absorbing,

$$\mathbb{P}(\text{flag} = 1) = \mathbb{P}(\tau_A \leq T) = p_T.$$

Amplitude estimation is then applied to estimate this probability.

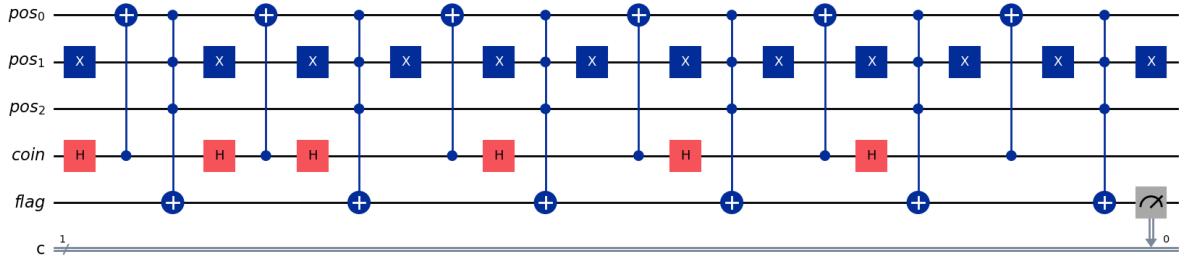


Figure 2: Quantum circuit implementing the coined quantum walk with position, coin, and flag registers.

2.3 Branching fault-tree model

Many reliability problems involve combinations of multiple conditional system states not a simple one-dimensional trajectory. We represent the problem as a binary tree. Each path through the tree represents a sequence of system states, and the rare event occurs when a path reaches a designated failure node.

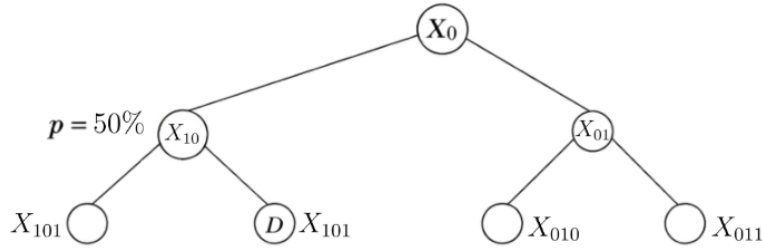


Figure 3: Branching fault-tree representation of system transitions.

2.4 Fault-tree quantum circuit

To encode the branching model, the walker position is generalized from a coordinate to a node in a graph. Branch qubits represent the choices along the tree, while ancilla qubits implement logical conditions corresponding to failure pathways.

Logical operations detect combinations of branch states that correspond to rare-event nodes. When such a state is detected, a flag qubit is activated. As in the one-dimensional model, amplitude estimation is then used to estimate the probability of reaching the flagged failure state.

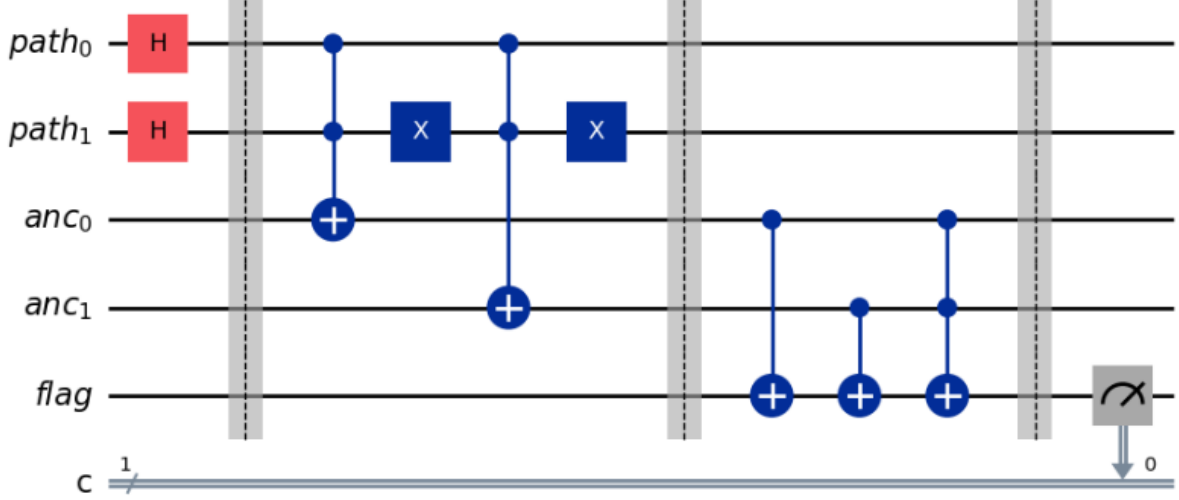


Figure 4: Quantum circuit implementing the branching fault-tree model with ancilla and flag qubits.

3 Results and Conclusion

Initial experiments were conducted on small problem instances where the walk depth and number of position qubits remain tractable. Simulations show that the quantum walk circuit correctly reproduces the expected probability distribution for short walks and identifies the absorbing-state probability. However, scaling the approach rapidly increases circuit depth, gate count, and qubit requirements. As a result, the current implementation is limited to small toy examples on classical simulators, and direct application on modern quantum hardware remains challenging.

A key finding is that circuit optimization strongly depends on the structure of the problem, such as a binary tree, graph, or one-dimensional walk. This conflicts with the objective of developing generic methods that avoid problem-specific tuning. Because circuit optimization for current quantum hardware is typically problem-specific, quantum methods may not provide an advantage over classical approaches.

Future work may address these limitations through improved circuit design and hybrid methods. Dynamic circuits could enable reuse of quantum registers during the walk, reducing qubit requirements. Circuit optimization may reduce gate count and the number of ancilla qubits. Hybrid classical–quantum approaches may also be explored, where classical simulation generates candidate trajectories and quantum routines estimate small probabilities more efficiently.

Availability of data and software code

Our code will be available at the following URL: <https://github.com/alonakos/Extremely-Rare-Event-Probability-Estimation>.

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References

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Supplemental Material

First-passage probability to an absorbing boundary in 1D

Suppose we have a one-dimensional classical random walker on $\{0, \dots, D\}$, with $D \geq 2$, that starts at position $X_0 = 0$. At each interior position $1 \leq X_s < D$, the walker moves:

$$\Delta X_s = \begin{cases} +1 & \text{with probability } p, \\ -1 & \text{with probability } 1 - p. \end{cases}$$

At 0, the walker reflects to 1 with probability 1. The state D is absorbing: once reached, the walker remains there.

We define the first-passage time

$$\tau = \inf\{t \geq 0 : X_t = D\},$$

which is the number of steps required for the walker to reach D for the first time. Our goal is to compute

$$p_T = \mathbb{P}(\tau \leq T),$$

the probability that the walker reaches D by a fixed deadline T . This provides a classical reference for the model discussed in the main text. We focus on deadlines close to D ; whether the event is rare also depends on D and p .

Step combinations and parity. Before absorption, every step changes the position by $+1$ or -1 . If the first arrival at D occurs after t steps, with k forward steps and $t - k$ backward steps, then

$$D = k - (t - k) = 2k - t.$$

Therefore,

$$k = \frac{t + D}{2}.$$

Because k must be an integer, first arrival is possible only when $t \geq D$ and t and D have the same parity. This restriction applies to the first-arrival time, not to occupancy of D after absorption.

Probability before absorption. Let

$$a_t(j) = \mathbb{P}(X_t = j, \tau > t), \quad 0 \leq j < D,$$

denote the probability of being at position j without having reached D . Initially,

$$a_0(0) = 1, \quad a_0(j) = 0 \quad (1 \leq j < D).$$

For convenience, set $a_t(D) = 0$. The probabilities evolve as

$$\begin{aligned} a_{t+1}(0) &= (1 - p)a_t(1), \\ a_{t+1}(1) &= a_t(0) + (1 - p)a_t(2), \\ a_{t+1}(j) &= p a_t(j - 1) + (1 - p)a_t(j + 1), \quad 2 \leq j < D. \end{aligned}$$

The last line is absent when $D = 2$. These recurrences account for reflection at 0 and removal of probability that reaches D .

First hit. A first arrival at D at time t occurs when the walker has survived at $D - 1$ until time $t - 1$ and then steps forward. Thus,

$$\mathbb{P}(\tau = t) = p a_{t-1}(D - 1), \quad t \geq 1.$$

This probability is zero when $t < D$ or the parity condition fails.

Cumulative probability before a deadline. Summing the first-arrival probabilities gives

$$\mathbb{P}(\tau \leq T) = \sum_{t=1}^T p a_{t-1}(D - 1) = 1 - \sum_{j=0}^{D-1} a_T(j).$$

Because D is absorbing,

$$\mathbb{P}(X_T = D) = \mathbb{P}(\tau \leq T).$$

When T is close to D . The earliest possible first arrival occurs at time D , followed by $D + 2$. The initial step from 0 is forced, so

$$\mathbb{P}(\tau \leq D) = p^{D-1}.$$

For arrival at time $D + 2$, there is exactly one backward step. A backward step from 1 returns the walker to 0, followed by a forced forward step. A backward step from any of $2, \dots, D - 1$ requires an additional probabilistic forward step. Therefore,

$$\mathbb{P}(\tau \leq D + 2) = p^{D-1} + p^{D-1}(1 - p)[1 + (D - 2)p].$$

The second term is not necessarily a small correction. For longer deadlines, the recurrences above give the exact classical first-passage probability.